

Rays

PHYLUM	Chordata
CLASS	Chondrichthyes
SUBCLASS	Elasmobranchi (part)
ORDERS	4
FAMILIES	17
SPECIES	About 650

Broad bodies and winglike fins distinguish rays from other cartilaginous fishes. Their flattened shape is generally an adaptation for life on the seabed, although several species, including mantas and eagle rays, swim in open water. Rays are found throughout the world but are particularly diverse in the tropics, while a group of egg-laying rays, called skates, are widespread in temperate waters. Most rays live along coasts, but some range far into deep-sea basins. Although some rays range between fresh- and saltwater estuaries, there are others that live exclusively in freshwater.

Anatomy

Rays have flat bodies; a long, thin tail; and large pectoral fins, which are used for propulsion. Larger species flap these fins like wings. Unlike sharks, they generally have a much smaller liver in proportion to their body, as most live on the sea bed and have no need for a buoyancy device. Most species are coloured to blend into the seabed, but they may also partially bury themselves in the sediment. As their mouth is on the underside, they breathe through openings called spiracles—one behind each eye. Water enters the spiracles and leaves through the gill openings. In addition to a covering of tiny placoid scales, many rays have large prickles or thornlike scales for protection and to provide grip while mating. While most rays are harmless, stingrays have a venomous barbed spine on the tail that can inflict a painful wound.

Feeding

All species of rays are carnivorous, and most are predators on small, bottom-dwelling fishes and

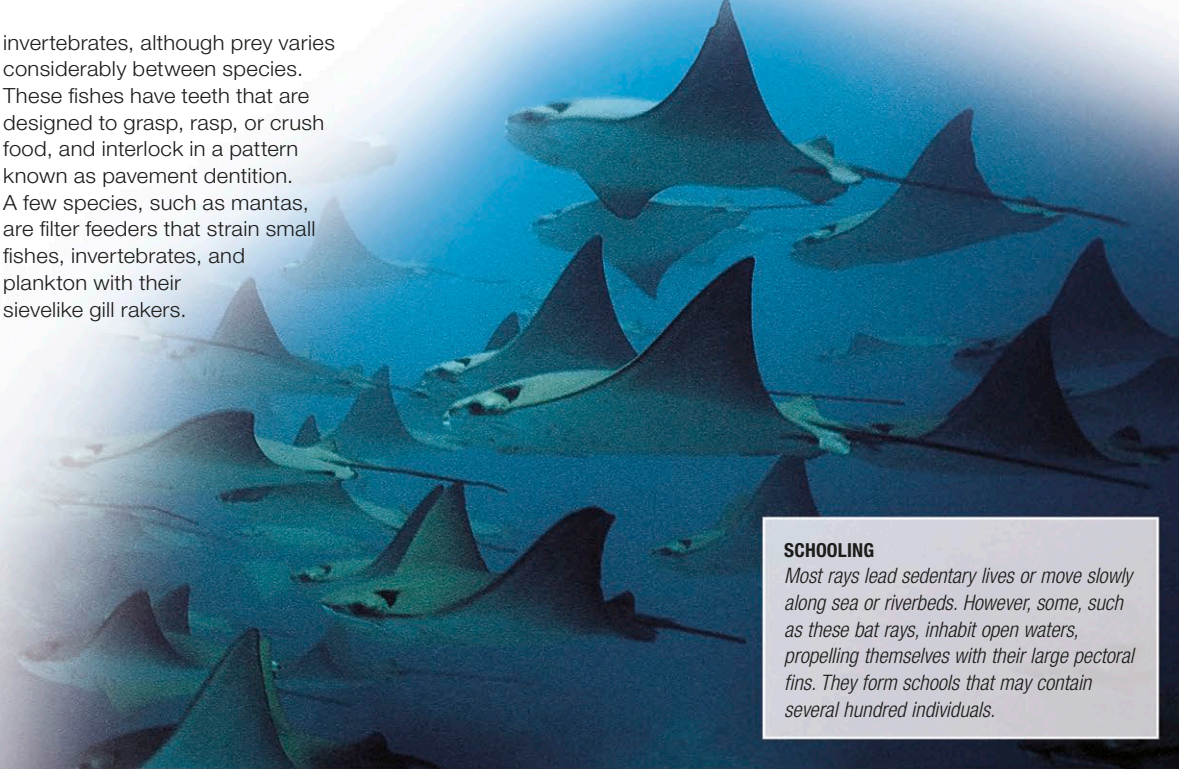
invertebrates, although prey varies considerably between species. These fishes have teeth that are designed to grasp, rasp, or crush food, and interlock in a pattern known as pavement dentition. A few species, such as mantas, are filter feeders that strain small fishes, invertebrates, and plankton with their sievelike gill rakers.

Reproduction

Like sharks, all rays mate and fertilization is internal. A large group of rays, particularly those commonly called skates, release large, leathery egg capsules containing one or more developing offspring. All other rays give birth to live young. The eggs hatch inside the female's body and the young live off their yolk sacs until large enough to be released. In some species, the embryos are further nourished by a fluid manufactured by the mother and produced by the membranes of the uterus.

CHIMERAS

Ratfishes, rabbitfishes, and their relatives—collectively known as chimeras—make up a small group of cartilaginous fishes that are distinct from sharks, skates, and rays. They form the class Holocephali, containing 52 species. Their soft, scaleless bodies are usually long and narrow, and they have a bulky head with large, iridescent eyes. Found in the waters of the Indian, Atlantic, and Pacific, they live at depths up to 26,400 ft (8,000 m).



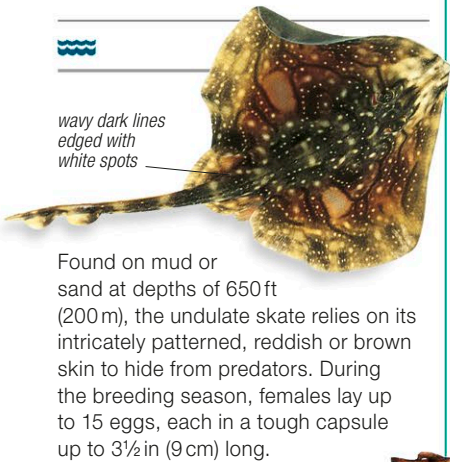
SCHOOLING

Most rays lead sedentary lives or move slowly along sea or riverbeds. However, some, such as these bat rays, inhabit open waters, propelling themselves with their large pectoral fins. They form schools that may contain several hundred individuals.

Raja undulata
Undulate skate

Length Up to 3¼ ft (1 m)
Weight 6½–8¼ lb (3–4 kg)
Breeding Oviparous
Status Endangered

Location E. Atlantic, Mediterranean

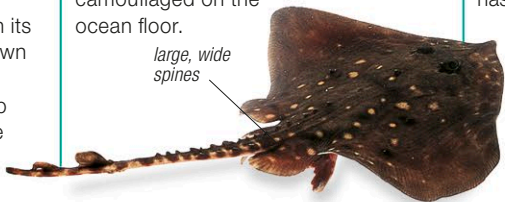


Raja clavata
Thornback skate

Length Up to 35 in (90 cm)
Weight 4½–8¼ lb (2–4 kg)
Breeding Oviparous
Status Near threatened

Location E. Atlantic, Mediterranean

The thornback skate occurs in marbled shades of brownish gray with a very varied pattern of lighter spots and blotches; its underside is white. Both sexes have big spines on their back—hence the common name of this species. As with most rays, it lies camouflaged on the ocean floor.

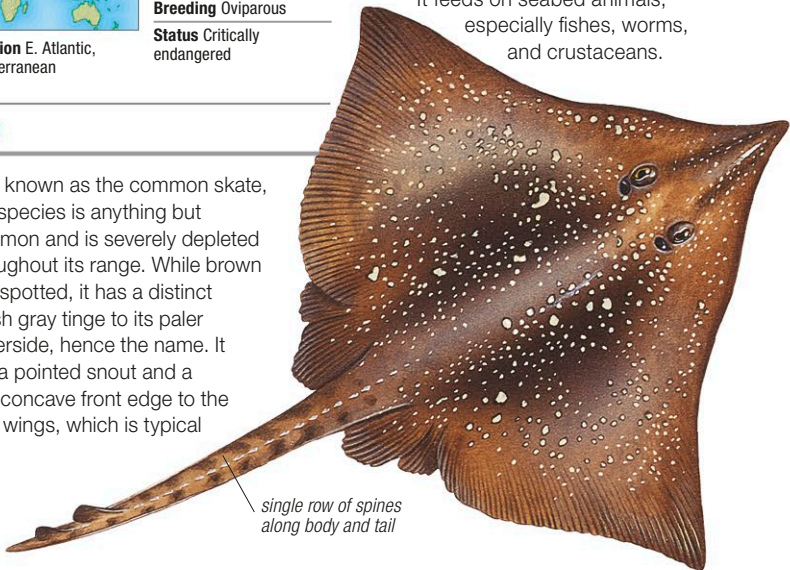


Dipturus batis
Blue skate

Length Up to 8¼ ft (2.5 m)
Weight 110–220 lb (50–100 kg)
Breeding Oviparous
Status Critically endangered

Location E. Atlantic, Mediterranean

Also known as the common skate, this species is anything but common and is severely depleted throughout its range. While brown and spotted, it has a distinct bluish gray tinge to its paler underside, hence the name. It has a pointed snout and a concave front edge to the wings, which is typical



of many skates. The female lays around 10 egg cases in pairs every year. These are very large and up to 8 in (20 cm) long, with a dark, thick, flexible shell and a pair of tendril-like thorns on either side. Unlike most other skates, the blue skate is active by day and night. It feeds on seabed animals, especially fishes, worms, and crustaceans.